BTS-III-11.14-0943 Reg.No. [ ] **B**

B.Tech. Degree III Semester Examination November 2014

IT/ME/EC/EB/EI 302 ELECTRICAL TECHNOLOGY

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) Derive emf equation of transformer.

(b) Write short notes on current transformer, potential transformer, distribution transformer and power transformer.

(c) Briefly explain motoring action and generating action.

(d) Briefly explain the necessity of starters in dc motors.

(e) Explain why synchronous motor is not self starting.

(f) Briefly explain rotating magnetic field.

(g) Briefly explain the concept of load dispatching.

(h) Explain the term, Corona.

PART B

(4 x 15 = 60)

II. A 20kva, 2500/250v, 50hz, single phase transformer gave the following test results: (15)

OC test (on L.V. side): 250v, 1.4A, 105w

SC test (on H.V side) : 104v, 8A, 320w

Obtain the following:

(i) the equivalent circuit referred to H.V. side

(ii) the equivalent circuit referred to L.V. side

(iii) regulation of transformer at half full load, 0.8 pf lagging load.

(iv) regulation of transformer at half full load 0.8pf. leading load.

(v) Secondary terminal voltage for case(iii)

(vi) Primary voltage required to be applied to get rated secondary voltage for case (iv)

(vii) Efficiency of transformer at ½ full load 0.8 pf lead.

(viii) The highest possible efficiency of the transform .

(ix) The load at which efficiency is 92% at upf.

OR

(P.T.O.)

III.	(a)	Explain OC test and SC test of a single phase transformer. Explain how the equivalent circuit of the transformer can be determined from these tests.	(8)
	(b)	A 600v/220v, single phase transformer takes no load current of 2A at a pf of 0.2 lag. The transformer supplies a load of 30A at a pf of 0.9 lag. Calculate the current drawn by the primary from the mains and primary power factor. Neglect the winding resistance and reactance.	(7)
IV.	(a)	Explain the process of voltage build up in a dc shunt generator.	(7)
	(b)	Explain the term armature reaction as applied to dc machine. How its effect can be reduced.	(8)

OR

V.	(a)	Explain the speed torque characteristics of DC series motor. List out some of its applications.	(7)
	(b)	In a 220v compound generator, the resistance of armature, shunt and series windings are 0.1Ω, 50Ω and 0.6Ω respectively. The load consists of 220 lamps each rated at 100w and 220v. Find the reduced emf when the machine is connected as (i) long shunt (ii) short shunt.	(8)
VI.	(a)	Derive emf equation of an alternator, also explain distribution factor and coil span factor.	(7)
	(b)	Find the no load phase and line voltage of a star connected 3 phase, 6 pole alternator which runs at 1200 rpm, having flux per pole of 0.2 wb sinusoidally distributed. Its stator has 54 slots having double layer windings. Each coil has 6 turns and the coil is chorded by one slot.	(8)

OR

VII.	(a)	What is synchronous condenser? Explain.	(7)
	(b)	Why single phase induction motor is not self starting? Briefly describe the different methods to make the motor start.	(8)
VIII.		Write short notes on: (i) Overhead lines and underground cables. (ii) Circuit breakers (iii) Electrical insulators (iv) Bus bars.	(15)

OR

IX.	(a)	Draw the elementary diagram of a typical substation and explain function of each block.	(10)
	(b)	What are the advantages and disadvantages of hydel power plant over other plants?	(5)

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